8. Select the criteria table in criteria range.
9. Click OK to display the matching records.

(*) Important Rule :
> Conditions in the Same Row = AND logic. (All conditions must be satisfied)

> Conditions in different Rows = OR logic (Any one condition can be satisfied).

* Solver :- Solver in Excel is a What-If Analysis tool used to find the best possible solution to a problem by changing selected values in cells while satisfying given constraints.

It is mainly used for optimization problems, such as maximizing profit, minimizing cost, & allocating resources efficiently.

(*) Solver works with three main components :-

1. Set Objective : The cell containing the formula you want to maximize, minimize, or set to a specific value.

2. By changing Variable Cells :- The cells Solver can change to find the best solution.

3. Subject to constraints :- The conditions or restrictions that Solver must follow.

(*) Types of Solver :-

(1) Max Solver => Used to maximize a value, such as total profit.
(2) Min Solver => Used to minimize a value, such as total shipping cost.
(3) Value Solver => Used to make a formula equal to a specific target value.

(1) Max Solver :- Max Solver is used to find the maximum possible value of an objective, such as profit, while satisfying all the given constraints.

(*) Example :- Product Profit Maximization.
~~Consider the following example from your image~~

| Product | Quantity | Profit Per Product | Overall Profit |
|---|---|---|---|
| Product A | 100 | 10 | 1,000 |
| Product B | 20 | 16 | 320 |
| Product C | 230 | 33 | 7,590 |
| Total | 350 | | 8,910 |

To Calculate the Overall profit for each product, multiply the Quantity by the profit per product.

To Calculate total profit :- Like (=SUM(C3:C5)).
& Same for total Quantity.

Constraints given:-

(1) At least 100 Units of Product A must be produced.

(2) At least 80 Units of Product B must be produced.

(3) At least 50 Units of Product C must be produced.

(4) The total production must equal 350 units.

(5) The Overall profit must be maximized.

(*) How to Use Max Solver :-

1. Open Solver → Go to the Data tab → Select Solver.
2. Set Objective . Choose Max.
3. By changing Variable Cells - These are the quantities Solver can change.
4. Add Constraints.
5. Select Solving Method :- Choose Simplex LP, since this is a linear optimization problem.
6. Click Solve.

(2) Min Solver :- Min Solver is used to find the minimum possible value of an objective, such as total cost, while satisfying the given constraints.

It is commonly used in transportation, production planning, & resource allocation problems.

(*) Example- Minimizing Total Shipping Costs

| Plants | QTR-1 | QTR-2 | QTR-3 | QTR-4 |
|---|---|---|---|---|
| Plant 1 | 92 | 40 | 92 | 92 |
| Plant 2 | 33 | 20 | 43 | 33 |
| Plant 3 | 55 | 20 | 55 | 55 |
| Total production | 180 | 80 | 190 | 180 |
| Warehouse demand | 180 | 80 | 190 | 180 |

| Plant | Shipping Cost per Unit |
|---|---|
| Plant 1 | 1.25 |
| Plant 2 | 1.84 |
| Plant 3 | 1.46 |

Output :-

| Plant | Total Units | Cost Per Unit | Total Shipping cost |
|---|---|---|---|
| Plant 1 | 316 | 1.25 | 395 |
| Plant 2 | 129 | 1.84 | 237.36 |
| plant 3 | 185 | 1.46 | 270.10 |
| Total | 630 | | 902.46 |

(*) Difference between Max Solver And Min Solver:-

| Max Solver | Min Solver |
|---|---|
| > Find the highest possible value. | > Find the lowest possible value. |
| > Maximize. | > Minimize. |
| > Profit Maximization. | > Cost Minimization. |
| > Maximize product profit. | > Minimize Shipping Cost. |
| > Product Quantities | > Production Allocation |
| > Simplex LP. | > Simplex LP. |